

Biomedical Image Processing

PYTHON REVIEW

Helena R. Torres

Python programming review

VARIABLES AND MATRICES

☐ Variable

- ☐ `x = 5` #variable called x with value 5

☐ Vector

- ☐ `x = [1,2,3,4]` #vector called x with four elements

☐ Matrix

- ☐ Python does not have a built-in type for matrices
- ☐ For matrices we can use the NumPy library
 - ☐ `import numpy as np` #import the NumPy library using the alias name *np*
`x = np.zeros((2,2))` #create a matrix x with 0s with two columns and two rows
`y = np.ones((2,2))` #create a matrix y with 1s with two columns and two rows
`z = np.array([[1, 2, 3], [3, 4, 5]])` #create a matrix z with values



Python programming review

MATRIX MANIPULATION

❑ Attribute a value to an element

- ❑ `x = np.zeros((3,2))` #create a matrix x with 0s with two columns and two rows
`x[0,0] = 10` #changing the first value of the matrix to 10
`x[0,1] = 5`
`x[1,1] = 3`

❑ Accessing elements

- ❑ `y = np.array([[1, 2, 3, 5], [3, 4, 5, 7], [8, 5, 2, 9]])` #create a matrix y
`y[0,3]` #element in the row 0 and column 3
`y[:,2]` #all elements in column 2
`y[1,:]` #all elements in row 1
`y[0:2,2]` #elements 0 to 2 in column 2
`y[1,0:4:2]` #elements 0, 2, and 4 in row 1 (intervale of 2 elements)

Python programming review

FUNCTION DEFINITION

- ❑ Programs that can be called with input parameters and that normally return an output parameter
- ❑ A function begins with the word **def**

```
def Value():
```

```
    x = 1
```

```
    return x
```

```
def Sum(y):
```

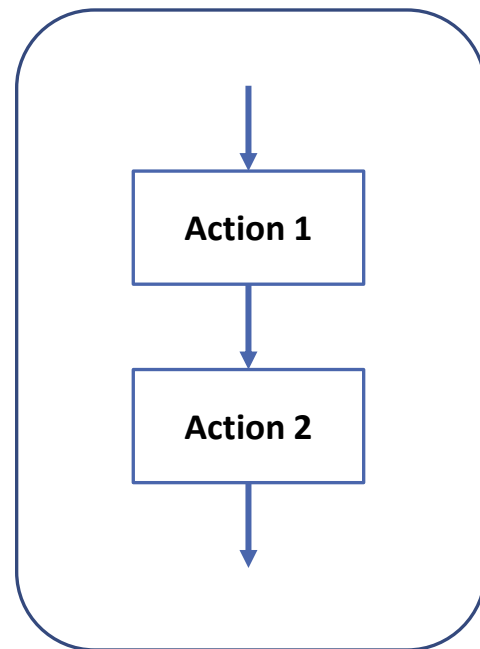
```
    x = 1
```

```
    z = x + y
```

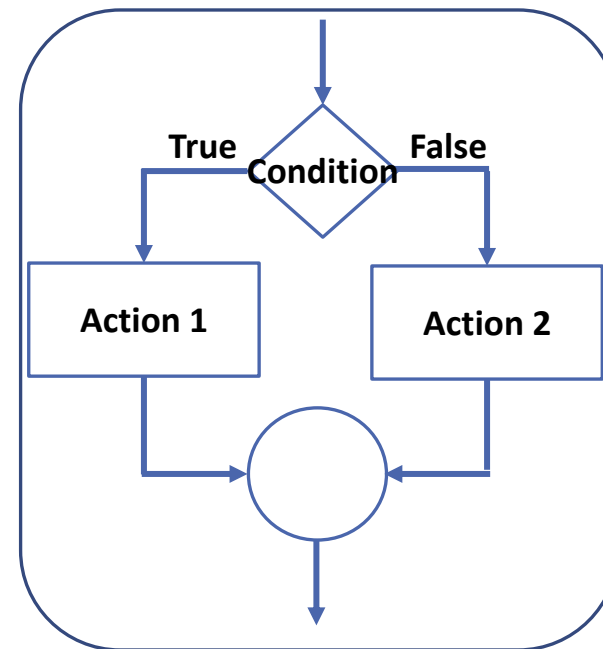
```
    return z
```

Flow control structures

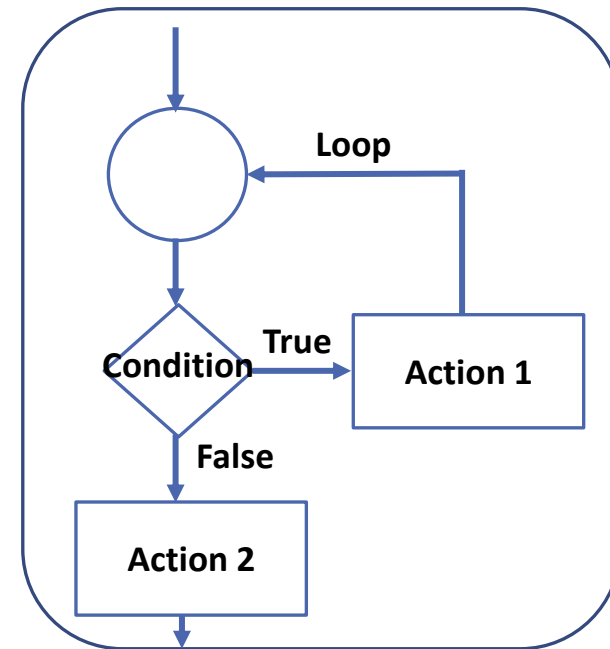
- ❑ In python, there are a large number of functions used to send messages and process data
- ❑ The flow of program execution should be easy to understand, and control structures can be used for that;
 - ❑ Sequential;
 - ❑ Conditional;
 - ❑ Iterative;



Sequential



Conditional



Iterative

Flow control structures

IF

- ❑ Check whether a given condition is true or not;
- ❑ If it is checked, the instruction within the block defined by the {} is executed;
- ❑ Otherwise:
 - ❑ it is not executed(case 1)
 - ❑ another operation is executed (case 2)
 - ❑ another operation is verified (case 3);

Case 1

```
if condition:  
    instruction
```

Case 2

```
if condition:  
    instruction1  
else:  
    instruction2
```

Case 3

```
if condition:  
    instruction1  
  
elif condition2:  
    instruction2  
  
else:  
    instruction3
```

Flow control structures

WHILE

- ❑ Execute a statement until a condition is no longer true;

```
while condition:  
    instruction
```

FOR

- ❑ It is used for iterating over a sequence, allowing code to be executed repeatedly
- ❑ The *range()* function returns a sequence of numbers, starting from 0 by default, and increments by 1 (by default), and ends at a specified number.

```
for i in range(Final):  
    instruction
```

```
for i in range(Start, Final, Increment):  
    instruction
```